Math 5320 Exam I 30 September 2013

In-Class

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** You may keep the exam sheet for your records.

Part A. Do any five (5) of the following problems:

1. (12) Let $z = 3 - \sqrt{3}\,i$ *and* $w = -2 + 2i$. Write in rectangular form, $a + bi$, and polar form, $r\,\text{cis}\,\theta$, each of the following:

   a. $\dfrac{w^2}{z^2}$ b. $(z + \overline{w})(\overline{z} + w)$

2. (12) Prove the following proposition: Let (X, d) be a metric space. Then, if $\{G_j : j \in J\}$ is a collection of open sets in X, J any indexing set, then $G = \cup\{G_j : j \in J\}$ is also open.

3. (12) Prove the following proposition: Let $f : (X, d) \to (\Omega, \rho)$ be a continuous function. If X is compact, then $f(X)$ is a compact subset of Ω.

4. (12) Prove the following proposition: Let $z, w \in \mathbb{C}$. Then, $zw = 0$ implies $z = 0$ or $w = 0$.

5. (12) Give examples of sequences $\{x_n\}$, $\{y_n\}$ in $\mathbb{C}$ such that

   a. $\lim x_n y_n = L$, $\lim x_n = M$, $\lim y_n$ does not exist

   b. $\lim x_n y_n = L$, $\lim x_n$ does not exist, $\lim y_n$ does not exist

   c. $\lim \dfrac{x_n}{y_n} = L$, $\lim x_n = M$, $\lim y_n = N$, $L \neq \dfrac{M}{N}$

6. (12) Prove the following proposition: Let (X, d) be a metric space. If X is compact, then every infinite subset of X has a limit point in X.

Part B. Do each of the following problems:

7. (15) Provide a counterexample to each of the following assertions:

a. In a metric space (X , d), if A, B are subsets of X with A closed in X, then $A \cap \overline{B} = \overline{A \cap B}$

b. In a metric space (X , d), for any $x \in X$ and $A \subset X$ we have: If $\operatorname{int} A \neq \varnothing$, then $d(x, \overline{\operatorname{int} A}) = d(x, A)$.

c. Let (X , d) be a metric space and let $f : X \to \mathbb{R}$ be continuous on X. If A is a open subset of X, then $f(A)$ is a open subset of $\mathbb{R}$.

d. Let (X, d), (Ω, ρ) be metric spaces. Let $f : X \to \Omega$. If f is continuous on X, then f is uniformly continuous on X.

e. Let (X, d) be a metic space. If X is complete, then X is compact.

8. (30) Classification Problem. Correctly identify whether the following subsets of $\mathbb{C}$ are: (a) open; (b) closed; (c) connected; (d) polygonally path connected; (e) compact; (f) complete; (g) bounded; (h) region. You do not need to provide a rationale for your classification. Fill out the classification information on the attached page, i.e.,

if the set possesses the property **mark the cell** in the table with Y (= yes)
if the set does not possess the property **do not mark** the cell in the table

A. $\{z = x + iy : x > 0, |y| < \frac{1}{|x|}\}$

B. $\{z = x + iy : \frac{x^2}{25} + \frac{y^2}{4} \leq 1\} \setminus B(0,3)$

C. $B(0,4) \setminus S$ where $S = \{z : |\operatorname{Re} z| \leq 2\}$

D. $\left\{\overline{B(0,1)} \setminus B(\tfrac{1}{2}, \tfrac{1}{2})\right\} \cap \{z : \operatorname{Re} z > 0\}$

E. $\bigcup_{n=1}^{\infty} l_n$, where each $l_n = [0, b_n]$ where $b_n = \frac{\pi}{2n} \operatorname{cis} \frac{\pi}{2n}$

Classification Table for Problem 8

	open	closed	connected	polygonally path connected	compact	complete	bounded	region
A								
B								
C								
D								
E								